

CATHERINE CHI CHUNG

usacchung@gmail.com || 408-442-6310 || San Francisco Bay Area
U.S. Work Authorization || github.com/usacchung

SUMMARY

Data-driven professional with 5+ years of experience in investment banking, leveraging analytical skills and cross-functional collaboration to solve complex financial problems. Columbia University graduate with a strong foundation in quantitative analysis and machine learning. Actively seeking roles in quantitative research, data science, or technology-driven finance. Intellectually curious and highly adaptable, with a hands-on approach to problem solving and a strong track record of delivering data-driven insights in collaborative environments.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, C, SQL, Excel VBA
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, Matplotlib, ggplot2
- **Data Science & ML:** Python, R, MATLAB (statistical modeling, machine learning, data wrangling)
- **Data Visualization:** Tableau, Power BI, R (ggplot2), Python (Matplotlib)
- **Tools & Platforms:** Git, Bloomberg, RStudio, Jupyter Notebook

QUANTITATIVE RESEARCH & DATA SCIENCE PROJECTS

Regime Detection Using Hidden Markov Models (HMM) 2025

- **Market regime modeling:** built a HMM to detect market regimes using engineered technical factors (e.g., RSI, returns, volatility), enabling pattern recognition in price dynamics
- **Unsupervised learning:** applied K-Means clustering and PCA/3D visualization for regime segmentation and interpretation across multiple market states
- **Signal evaluation:** conducted feature importance analysis with Spearman and Pearson correlations to assess predictive strength of alpha signals
- **Full Python pipeline:** developed end-to-end data pipeline (data wrangling, feature engineering, modeling, and visualization) using real market data

WORK EXPERIENCE

Leotek Electronics USA (via Redoak Tech), San Jose, CA Mar 2024–Feb 2025

Business Analyst

- **Statistical modeling & data integration:** built SQL/Python pipeline combining production, repair, and feedback data; applied logistic regression and tree-based models to reduce defect rate by 18% and improve yield by 12%

Guotai Junan Securities, Shanghai, China Mar 2019–Mar 2023

Investment Banking Associate (including risk team for 2 years)

- **Non-performing loans modeling:** compared the performance of a wide set of regression techniques and ML algorithms for predicting recovery rates on non-performing loans, using banks' database and a self-built database from public source with Excel, SQL and R
- **ML modeling:** designed a cashflow forecasting model using ML algorithms (random forest and logistic regression) with R to reduce a finance company's debt costs by around \$1.5 million; communicated technical findings to both technical and non-technical stakeholders
- **Securities pricing:** collaborated with product managers to perform pricing analysis; developed SQL queries to extract data from database, resulting in a 30% time reduction; supported investment by visualizing large datasets using Python packages, such as Numpy and matplotlib; presented the findings to stakeholders and offered strategic pricing advice
- **Risk analysis:** evaluated credit risk, including qualitative and quantitative approaches; analyzed risk issues such as collateral analysis and macroeconomic stress test scenarios

EDUCATION

- **Columbia University** | M.S. Financial Engineering | New York, NY | 2018
- **National Tsing Hua University** | B.S. Quantitative Finance | Hsinchu, Taiwan | 2013

LEADERSHIP & VOLUNTEERING

- **Columbia Alumni Rep Committee,** Interviewer | 2020
- **Young Presidents' Organization,** Event Assistant | 2016
- **Team Captain,** NTHU Varsity Table Tennis | 2010–2012